

# BUCHEL, Germany

WMO: 106130

Lat: 50.1747N

Lon: 7.0594E

Elev: 486

StdP: 95.62

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -9.3       | -7.4 | -12.2                         | 1.4 | -7.0 | -10.0 | 1.7 | -5.6 | 11.7                  | 5.5  | 10.5 | 4.7  | 3.7                   | 60   | 0.572 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.6                    | 28.2            | 18.3 | 26.1 | 17.5 | 24.2 | 16.7 | 19.3                | 25.9 | 18.4 | 24.4 | 17.5 | 22.9 | 3.4                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 17.1                            | 12.9 | 21.1 | 16.1 | 12.2 | 20.3 | 15.3 | 11.5 | 19.6 | 56.8          | 26.0 | 53.9 | 24.1 | 51.2 | 22.8 | 22.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 9.1               | 8.0  | 7.1 | -11.6                      | 31.4 | 3.0                | 2.1 | -13.7                                              | 32.9 | -15.5      | 34.1 | -17.2      | 35.3 | -19.3      | 36.8 |    |    |
|                   |      |     | -11.8                      | 21.1 | 2.8                | 0.9 | -13.8                                              | 21.8 | -15.4      | 22.3 | -17.0      | 22.8 | -19.1      | 23.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 8.7    | 0.2   | 1.2   | 4.5   | 8.4   | 12.1  | 15.4  | 17.5  | 17.0  | 13.3  | 9.1   | 4.4   | 1.2   |  |  |
|                                                                   | DBStd       | 7.22   | 4.14  | 4.09  | 4.01  | 4.21  | 3.94  | 3.84  | 3.80  | 3.65  | 3.31  | 3.59  | 3.63  | 4.05  |  |  |
|                                                                   | HDD10.0     | 1340   | 304   | 247   | 175   | 79    | 22    | 2     | 0     | 0     | 6     | 60    | 171   | 274   |  |  |
|                                                                   | HDD18.3     | 3594   | 562   | 481   | 429   | 297   | 196   | 104   | 63    | 69    | 156   | 286   | 418   | 532   |  |  |
|                                                                   | CDD10.0     | 877    | 0     | 0     | 4     | 32    | 88    | 163   | 232   | 218   | 104   | 33    | 2     | 0     |  |  |
|                                                                   | CDD18.3     | 89     | 0     | 0     | 0     | 0     | 3     | 16    | 37    | 29    | 4     | 0     | 0     | 0     |  |  |
|                                                                   | CDH23.3     | 618    | 0     | 0     | 0     | 1     | 21    | 106   | 258   | 210   | 22    | 0     | 0     | 0     |  |  |
|                                                                   | CDH26.7     | 141    | 0     | 0     | 0     | 0     | 2     | 18    | 61    | 58    | 2     | 0     | 0     | 0     |  |  |
| Wind                                                              | WSAvg       | 3.5    | 4.0   | 4.0   | 3.9   | 3.5   | 3.4   | 3.2   | 3.2   | 3.1   | 3.2   | 3.5   | 3.6   | 3.9   |  |  |
| Precipitation                                                     | PrecAvg     | 725    | 62    | 53    | 52    | 46    | 63    | 68    | 67    | 62    | 59    | 61    | 62    | 69    |  |  |
|                                                                   | PrecMax     | 942    | 183   | 149   | 132   | 91    | 120   | 124   | 179   | 134   | 136   | 158   | 117   | 198   |  |  |
|                                                                   | PrecMin     | 553    | 11    | 6     | 6     | 0     | 18    | 19    | 29    | 12    | 14    | 21    | 4     | 13    |  |  |
|                                                                   | PrecStd     | 101    | 34    | 32    | 30    | 22    | 27    | 30    | 32    | 32    | 33    | 30    | 26    | 37    |  |  |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 10.1  | 11.9  | 17.1  | 22.3  | 25.8  | 29.1  | 31.1  | 31.8  | 26.1  | 20.4  | 14.0  | 10.2  |  |  |
|                                                                   |             | MCWB   | 8.6   | 6.9   | 9.8   | 13.1  | 16.9  | 18.8  | 18.8  | 19.3  | 17.2  | 15.0  | 11.8  | 9.1   |  |  |
|                                                                   | 2%          | DB     | 8.7   | 9.2   | 14.2  | 19.9  | 22.9  | 26.2  | 28.3  | 28.0  | 23.0  | 17.6  | 11.9  | 9.2   |  |  |
|                                                                   |             | MCWB   | 7.7   | 6.5   | 8.5   | 11.8  | 15.3  | 17.9  | 18.2  | 18.3  | 16.3  | 13.9  | 10.2  | 8.3   |  |  |
|                                                                   | 5%          | DB     | 7.1   | 7.9   | 12.0  | 17.2  | 20.8  | 24.1  | 26.1  | 25.5  | 20.7  | 15.5  | 10.5  | 8.0   |  |  |
|                                                                   |             | MCWB   | 6.3   | 6.3   | 7.8   | 10.8  | 13.8  | 16.8  | 17.6  | 17.1  | 15.4  | 13.1  | 9.3   | 7.4   |  |  |
|                                                                   | 10%         | DB     | 5.7   | 6.7   | 10.1  | 15.1  | 18.8  | 22.1  | 24.1  | 23.2  | 18.8  | 14.0  | 9.1   | 6.7   |  |  |
|                                                                   |             | MCWB   | 5.0   | 5.5   | 7.0   | 9.8   | 13.0  | 15.9  | 16.7  | 16.5  | 14.5  | 12.2  | 8.3   | 6.2   |  |  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 9.4   | 8.5   | 10.8  | 14.1  | 18.1  | 20.6  | 20.4  | 20.2  | 18.3  | 15.7  | 12.2  | 9.7   |  |  |
|                                                                   |             | MCDB   | 9.7   | 9.7   | 14.6  | 20.2  | 24.3  | 26.4  | 28.0  | 27.8  | 23.4  | 18.7  | 13.3  | 10.0  |  |  |
|                                                                   | 2%          | WB     | 7.7   | 7.4   | 9.6   | 12.6  | 16.2  | 18.7  | 19.2  | 19.0  | 17.0  | 14.4  | 10.6  | 8.5   |  |  |
|                                                                   |             | MCDB   | 8.3   | 8.5   | 12.7  | 18.5  | 21.4  | 24.5  | 25.9  | 26.2  | 21.9  | 16.7  | 11.6  | 9.0   |  |  |
|                                                                   | 5%          | WB     | 6.3   | 6.4   | 8.5   | 11.2  | 14.7  | 17.6  | 18.3  | 18.0  | 15.9  | 13.4  | 9.5   | 7.3   |  |  |
|                                                                   |             | MCDB   | 7.0   | 7.5   | 11.2  | 15.9  | 19.5  | 23.0  | 24.4  | 23.8  | 19.6  | 15.2  | 10.4  | 7.8   |  |  |
|                                                                   | 10%         | WB     | 4.9   | 5.4   | 7.4   | 10.2  | 13.4  | 16.4  | 17.4  | 17.1  | 14.8  | 12.3  | 8.3   | 6.0   |  |  |
|                                                                   |             | MCDB   | 5.4   | 6.5   | 9.7   | 14.6  | 17.7  | 21.0  | 22.8  | 22.0  | 17.9  | 13.8  | 9.1   | 6.4   |  |  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 4.0   | 4.9   | 6.5   | 7.9   | 8.1   | 8.6   | 8.6   | 8.4   | 7.3   | 5.8   | 4.2   | 3.8   |  |  |
|                                                                   |             | MCDBR  | 5.3   | 6.6   | 9.3   | 10.6  | 10.5  | 11.2  | 11.2  | 11.3  | 9.8   | 7.8   | 5.5   | 4.4   |  |  |
|                                                                   |             | MCWBR  | 4.7   | 4.7   | 5.3   | 5.4   | 5.5   | 5.7   | 4.6   | 4.9   | 4.9   | 4.8   | 4.2   | 4.0   |  |  |
|                                                                   | 5% WB       | MCDBR  | 4.9   | 5.6   | 7.8   | 9.5   | 9.7   | 10.3  | 10.1  | 9.7   | 8.7   | 7.0   | 5.1   | 4.2   |  |  |
|                                                                   |             | MCWBR  | 4.7   | 4.5   | 5.0   | 5.1   | 5.4   | 5.7   | 4.7   | 4.8   | 4.8   | 4.7   | 4.3   | 4.0   |  |  |
| Clear-Sky Solar Irradiance                                        | taub        | 0.301  | 0.326 | 0.357 | 0.386 | 0.392 | 0.395 | 0.395 | 0.395 | 0.390 | 0.363 | 0.347 | 0.322 | 0.299 |  |  |
|                                                                   | taud        | 2.441  | 2.390 | 2.322 | 2.279 | 2.289 | 2.299 | 2.291 | 2.333 | 2.392 | 2.431 | 2.439 | 2.429 | 2.429 |  |  |
|                                                                   | Ebn at Noon | 746    | 811   | 843   | 856   | 865   | 863   | 856   | 842   | 829   | 775   | 707   | 689   | 689   |  |  |
|                                                                   | Edh at Noon | 61     | 82    | 104   | 121   | 125   | 125   | 124   | 113   | 96    | 78    | 61    | 54    | 54    |  |  |
| All-Sky Solar Radiation                                           | RadAvg      | 0.78   | 1.48  | 2.63  | 4.17  | 4.91  | 5.43  | 5.14  | 4.41  | 3.29  | 1.91  | 0.88  | 0.60  | 0.60  |  |  |
|                                                                   | RadStd      | 0.12   | 0.28  | 0.40  | 0.50  | 0.48  | 0.51  | 0.58  | 0.44  | 0.41  | 0.19  | 0.11  | 0.09  | 0.09  |  |  |

## Historical Trends

|                         |       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)                     | (e)   | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| Station Only            | +0.46 | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | -163        | +66     | N/A     | N/A     |
| Regional (62 neighbors) | N/A   | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page